

COMP9727 Recommender Systems



ASKreddit

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Problem

Reddit, as of 2025, has

- Over 1.1 billion monthly active users and 97 million daily active users
- More than 100,000 active subreddits
- 11 million posts per month, over 350,000 per day

How to best spend your limited (reddit) time the way you want to ?!?!

- World news? Best food? Connect with other users?

Our job is to create a reddit recommender system by analysing the voting history of users

Reddit's recommender

Reddit has an existing recommender¹, based on

- Content-related information, upvotes, downvotes and comments of content
- User activity, the subreddit the user visits and are subscribed to, user upvotes and downvotes and topics of interest selected at account creation
- Account age, newer accounts would get a broader range of recommendations rather than focused on subscriptions
- Location, recommendations may be based on the user's location settings

¹ reddit, 2025, Reddit's Approach to Content Recommendations, [https://support.reddithelp.com/hc/en-us/articles/](https://support.reddithelp.com/hc/en-us/articles/23511859482388-Reddit-s-Approach-to-Content-Recommendations)

Data

Motivational data

- Huge collection of Reddit Votes¹ on Kaggle, contains over 44 million upvotes and downvotes data cast between 2007 and 2020
- Their voting history reveals implicit preferences, interests and biases, we can use these information to personalize and rank content
- This dataset does not contain the contents of the posts that the votes apply to, so actual posts were downloaded using PRAW²
- Due to the massive dataset, our recommender focuses on just 1 subreddit r/Showerthoughts
 - This was chosen because it was a predominately text subreddit, and we excluded posts with images or videos

¹ Joseph Leake 2000, Huge Collection of Reddit Votes, Kaggle, <https://www.kaggle.com/datasets/josephleake/huge-collection-of-reddit-votes>

² PRAW: The Python Reddit API Wrapper, <https://praw.readthedocs.io/en/stable/index.html>

Data challenges

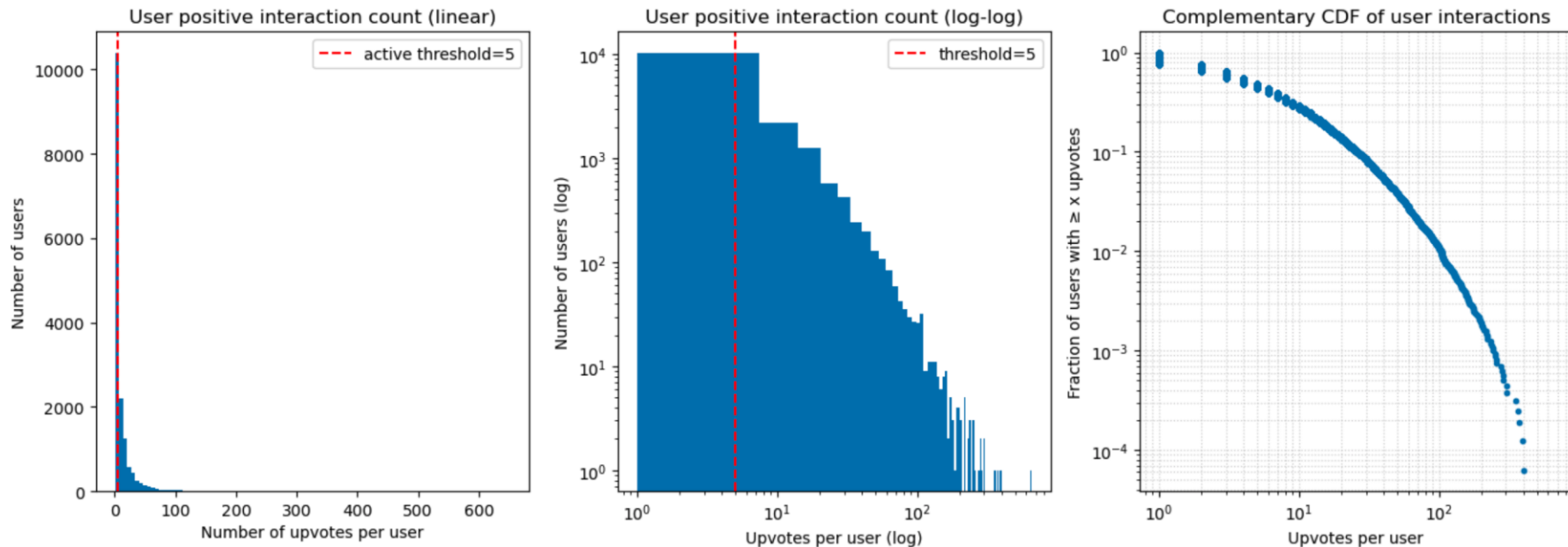
Some of the data challenges

- Not all users make their voting history public,
 - the data shows how many upvotes and downvotes there are on a post, but
 - we only know the individual users that upvoted or downvoted for a fraction of the votes
- We also do not know the browsing history of each user, and since it would not be practical
 - For the majority of users, we don't know which posts they have seen and have voted on
 - For a small fraction of users that we know which posts they have voted on, we do not know which posts they have seen (and not voted on)

Sample data

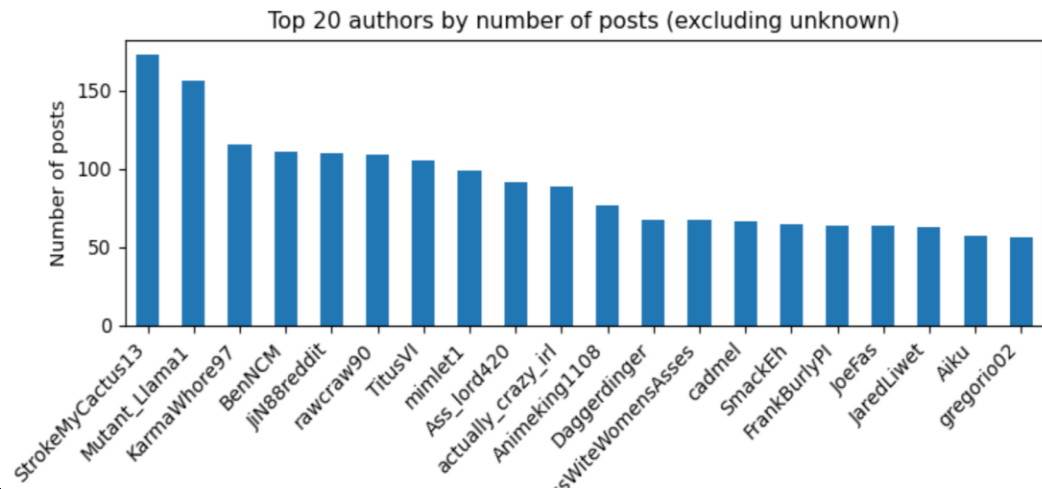
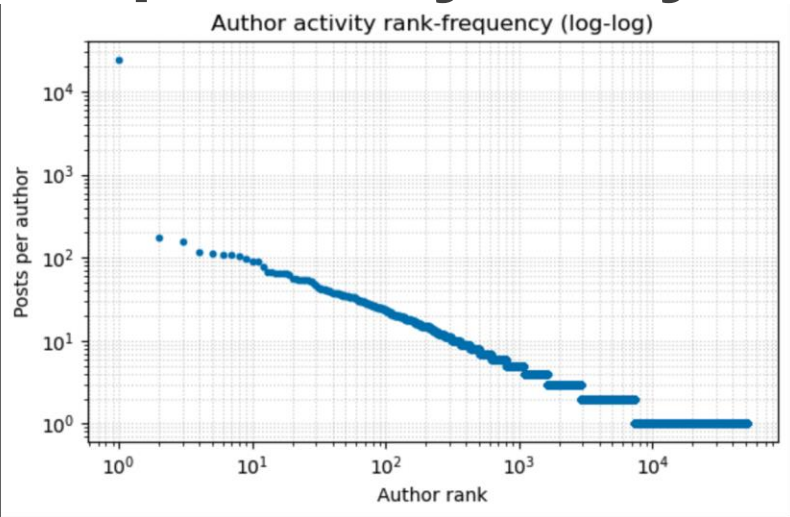
	submission_id	title	selftext	# num_comments	# num_unique_comment...	ups
0	t3_ea85qk	You might have seen a random pers	Missing value	0.0	0.0	23
1	t3_201knk	If you have two arms, then you have	Because just one person with one ar	0.0	0.0	24
2	t3_7wa7jw	Statistically speaking, you have a gre	Crazy, huh?	0.0	0.0	3
3	t3_cnyfjq	Gay genes are transferred to the unb	Missing value	0.0	0.0	0
4	t3_c1bgf9	A woodpecker has to have headache	Missing value	0.0	0.0	3
5	t3_dv1u7e	Jake from adventure Time could mak	Missing value	0.0	0.0	111
6	t3_3a4puu	When I die I want to be buried at a f	Missing value	0.0	0.0	1391
7	t3_d3jil3	When vibrations in the air hit your ea	Missing value	0.0	0.0	51
8	t3_bv6yox	If we survive, there may eventually b	Missing value	0.0	0.0	7
9	t3_8hzlif	If a couple of people are asses to yo	Missing value	0.0	0.0	33
10	t3_cqvygz	Kids songs can be so creepy	"It's raining, it's pouring. The old mar	0.0	0.0	15
11	t3_37kx9h	For me, all Reddit posts are NSFW. B	Missing value	0.0	0.0	914
12	t3_90zdv6	Having slow internet is infinitely wor	Missing value	0.0	0.0	355
13	t3_ep4qu8	If you think about it, there's probably	Missing value	0.0	0.0	0
14	t3_e43i98	There are two types of infinity. One t	Missing value	0.0	0.0	506
15	t3_dyozi4	Being alone with your love on an isla	Imagine the main character in a mov	0.0	0.0	3
16	t3_3914x8	"Instagram" would be a great name	Missing value	0.0	0.0	192
17	t3_a4j21c	Wishing you weren't born is basically	Missing value	0.0	0.0	2
18	t3_atgoz6	Family Guy have Referenced Robot C	Missing value	0.0	0.0	8
19	t3_3vmmtj	People always worry about burglars	Missing value	0.0	0.0	12
20	t3_aljnv4	The most unrealistic thing about mo	Missing value	0.0	0.0	101115

Exploratory analysis of the dataset



The interaction is extremely unbalanced. The vast majority of users have very few positive feedbacks (upvotes), while a small number of users are highly active (long-tail feature). From the CDF, only a very small proportion of users contributed a large amount of interaction.

Exploratory analysis of the dataset



The number of posts made by authors also follows a typical power-law distribution: a few authors are highly active, while the majority contribute only a small number of posts.

The number of "unknown" authors is extremely large

Methodology

The team approached the problem without a pre-conceived view on what approach would work well in this situation. We explored a number of recommender system approaches, and these are detailed in the following pages

- Neural network classification
- LLM embeddings
- SVM
- Vector Negation with word embeddings
- Sentence-BERT embeddings

Methodology – Neural network

A standard multiple layer perceptron (MLP) neural network was used as one of the approaches in designing our recommender. This is a fairly simply approach where the MLP was used to predict user upvotes at the user level, that is, each "upvote" out of a user's voting history is a "1" and an MLP was training to classify a post as a 1 or 0 based on if the user is likely to upvote.

Final model's parameters

- MLP from scikit-learn MLPClassifier
- 2 hidden layers of 128 and 64 nodes
- SMOTE from imblearn used to address the imbalanced nature of the data
- Default loss function (log-loss, binary entrophy) and solver (adam)
- Due to sparse data, dimension reduction techniques were tried

Methodology – LLM

We moved beyond traditional keyword-based methods and employed a Large Language Model (LLM) as a feature encoder.

- **Model Selection** – “all-mpnet-base-v2” * from sentence-transformers
- **Text Encoding** – “title” and the “selftext” from the posts data
- **Embedding Generation** – 768-dimensional vector representation for each entry
- **Efficient Storage** – saved the vectors and IDs into a compressed NumPy file (.npz)
- **Build User Profile** - a time-decay weighted average of the voted embeddings
- **Rank Recommendations** - measure the distance between user profile and candidates

* https://www.sbert.net/docs/sentence_transformer/pretrained_models.html

Methodology – Support Vector Machine

- Vectorised each post content (title + self text)
- User data would contain lists of post they either upvoted or downvoted
- labeled all upvotes by a user as 1 and downvotes as 0
- Trained a binary SVM classifier (`sklearn.svm.SVC`) per user
- The SVM learns a hyperplane to separate upvoted from downvoted content
 - Recommend based on distance from the hyperplane

Methodology – sentence-BERT + LightFM

- Active user/item filtering & per-user temporal holdout
- Content representation: SBERT embeddings, semantics, LDA topics distributions, sentiment scores, normalized metadata and author buckets
- Collaborative signal: user–post upvotes
- Hybrid ranking model: LightFM (interaction + item features) with hyperparameter tuning
- Evaluation: nDCG@200

Methodology – Vector Negation with Word2Vec

Let $a, b \in V$. Then a NOT b is represented by the vector

$$a \text{ NOT } b \equiv a - \frac{a \cdot b}{|b|^2} b$$

where $|b|^2 = b \cdot b$ is the modulus of b .

- source: Widdows, 2003.

- Extract keywords from posts using YAKE! (Campos, et al., 2020.)
- Use Byte-pair Encoding implementation from Heinzerling, et al., 2018.
- Modelling user profile with (pos_1 OR pos_2 ... OR pos_n) NOT (neg_1 OR neg_2 ... OR neg_n)
- Normalize upvotes(likes) with minmax, then return a weighted sum of similarity and upvotes.

Experimental Design

Posts in the r/Showerthoughts were split into 80% training data and 20% validation data

In the training set, we identified the **top 20 users** with the most votes (upvotes + downvotes), user profiles were created for these users and they are our “test subjects”.

For the **20% validation set** (roughly 19,000 posts), we used each recommender approach to generate **200 personalised post recommendations** per user using the **most optimised model version** of each method.

Evaluation – metrics

To compare the performance of the recommenders approaches undertaken, we have used the following metrics to compare and evaluate their performances

- Recall@N, Coverage, Precision@N -> not that good
- NDCG (Normalized Discounted Cumulative Gain)
 - We care not just about what's recommended, but where it appears in the list.
 - NDCG directly evaluates this: it gives higher weight to relevant items shown higher up, which reflects real user experience.

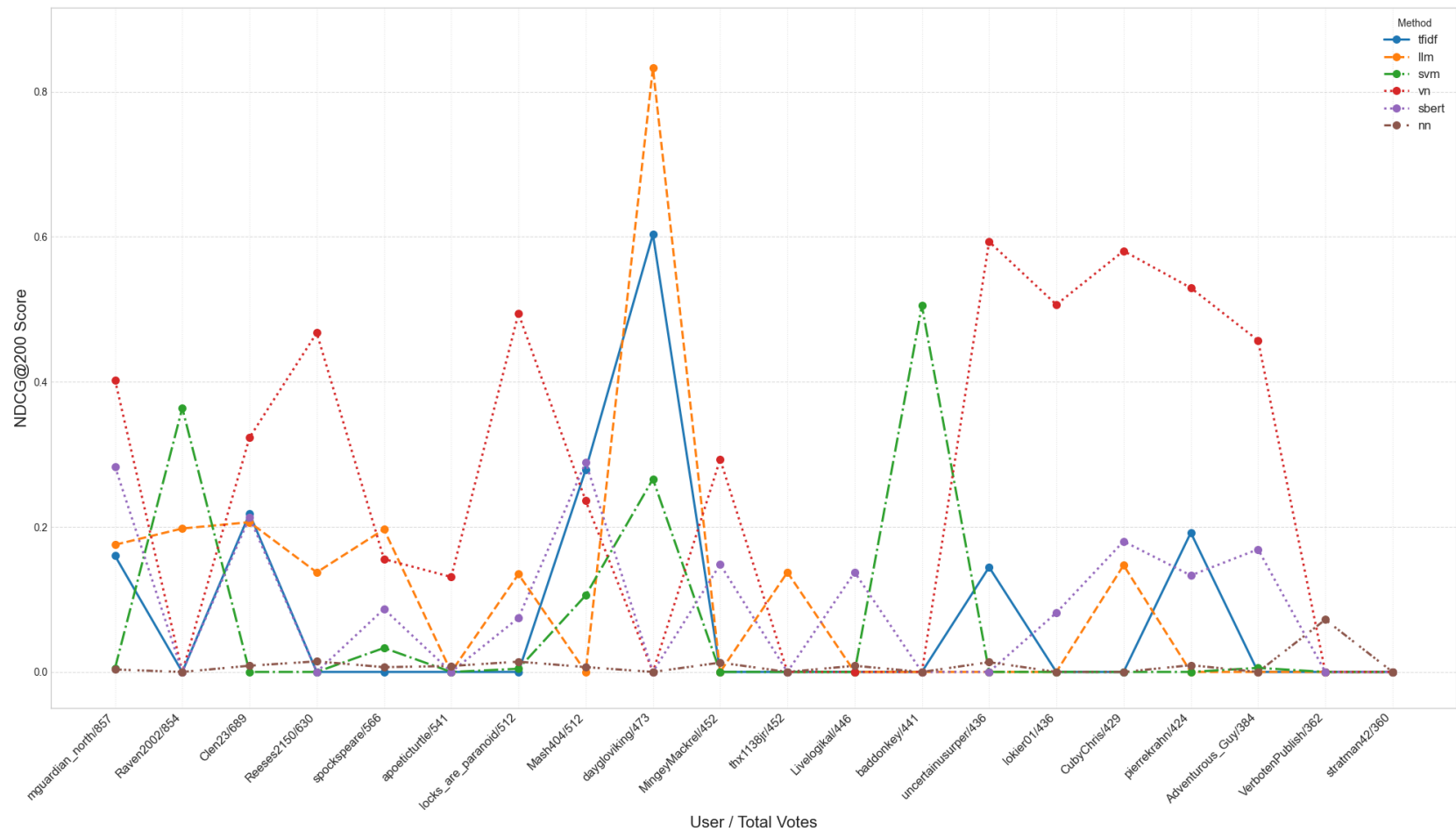
Our test data is extremely sparse/imbalanced

- most users have not sufficient enough votes. This makes metrics like Precision or Recall unreliable
- they often score 0 not because the recommender failed, but because there's no signal to match.

Evaluation – results

Approach	NDCG@200
<u>TF-IDF (baseline)</u>	<u>0.0798</u>
Neural Network	0.0089
SVM	0.0645
Sentence-BERT	0.0898
LLM	0.1083
Vector Negation	0.2585

Per-User NDCG@200 Comparison Across All Methods



Reflections and further research

This was a difficult problem, due to the data censoring, and it may be difficult to compare its performance to the default reddit recommender.

Different recommender approaches performed differently for different users, this creates opportunity for an ensemble method

The team faced a number of challenges whilst designing and developing our recommender approaches, whilst we could not fit into this project, the following would be worthwhile avenues for further research

- Multiple subreddits
- Online learning to update models based on user activities
- Use actual test subjects rather than historical voting data
- Extend the method to images and videos

Thank you and Questions?